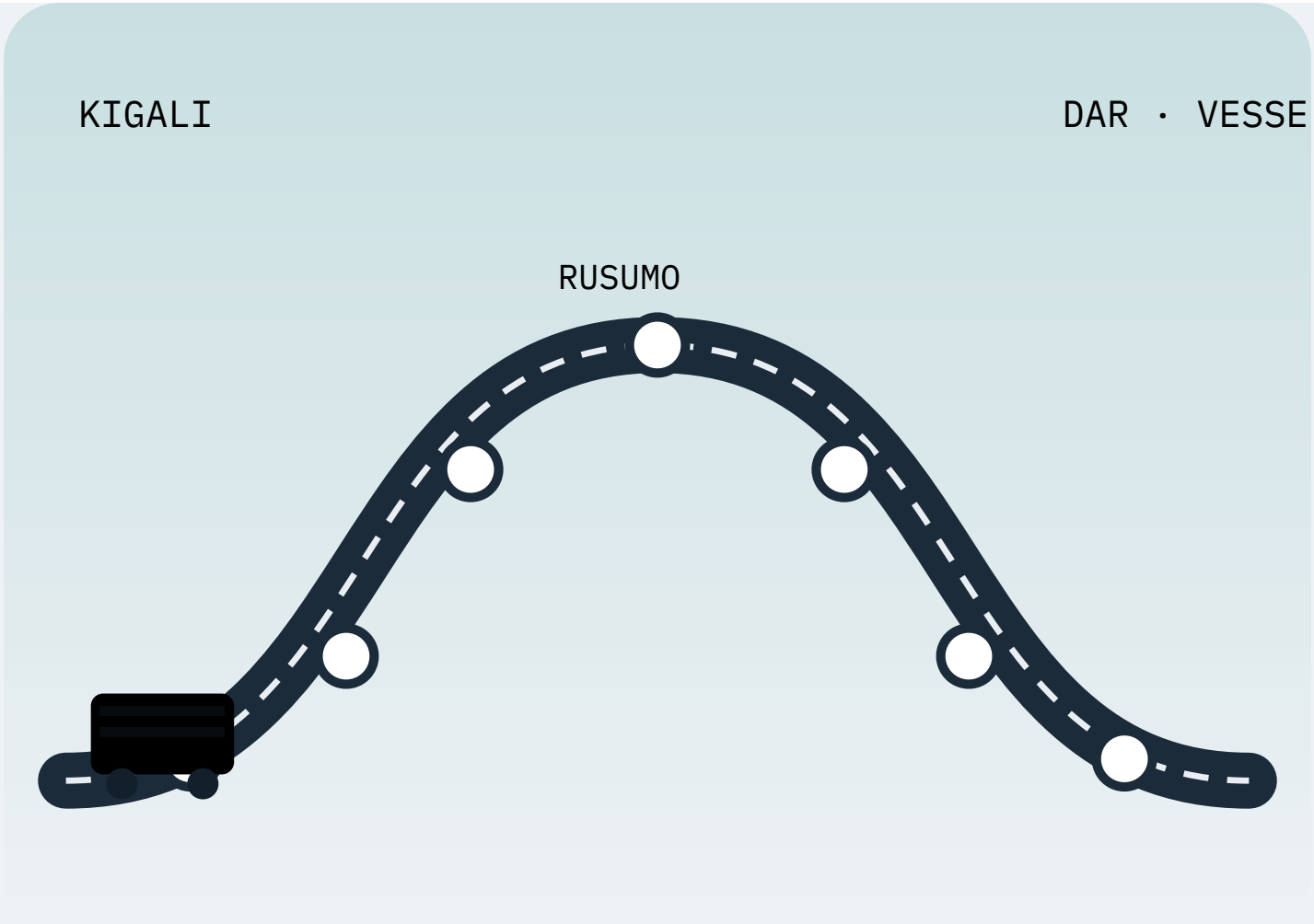
CORRIDOR MIS · BUILD ROAD · ONE FEATURE AT A TIME

Twelve slices. Each one used by real people before the next is started.

The system in the analysis is large. The way to build it without drowning a developer is to treat it like the corridor itself: one station at a time, a test at each gate, and the container does not move on until the gate is passed. Every slice below says what gets built, which menus of the mockup it switches on, the runs the developer walks through with real OLIU data, the test that must pass, and what the Managing Director sees when it does. The owner portal, for you, comes in two stages at the end, when there is a second and a third account to justify it.

> Rule of the road: a slice is finished when staff have used it on live bookings for a week and the test passed. Not when the code is written.

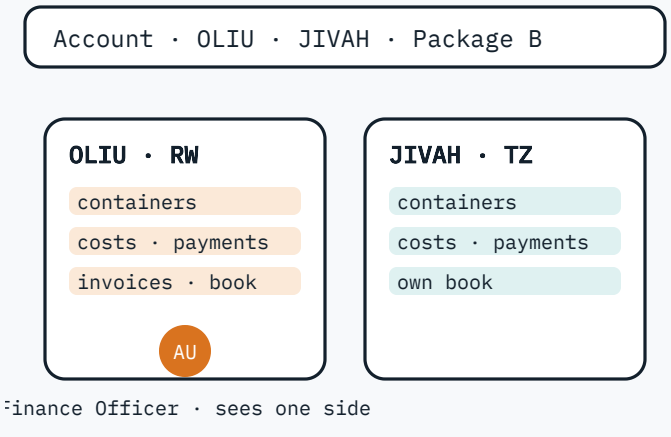


1

Accounts, companies, roles

SPINE · 2 WEEKS

Everything that comes later is scoped by two questions: which account, which company. Build that first, with nothing inside it, and the rest of the system can never leak across companies by accident.



WHAT GETS BUILT

- Account
- Company (country, currency, TIN, banks)
- Package A or B
- User + six roles
- Role attached to one company or all
- OTP to department mailbox
- Audit trail table

MENUS SWITCHED ON IN THE TENANT PORTAL

- **Settings** → User management
- **Settings** → Company & subscription: companies, branch or partner, who bills, hand-over point
- **Login** → OTP to the department mailbox

OWNER PORTAL, SAME STAGE

- **Owner** → Create account: the set-up questions, nothing else

The owner portal at this stage is one form you use yourself to create OLIU · JIVAH. No tenant list, no billing. It is the same set-up screen the tenant later sees under Settings.

RUNS THE DEVELOPER WALKS THROUGH

Run 1	**Create the account.** Answer the set-up questions: two companies, OLIU bills the client, hand-over at Rusumo. Nothing else is asked yet.
Run 2	**Invite twelve people.** Each gets one role and one company; the MD gets all. OTP arrives at finance@, ops@, management@.
Run 3	**Sign in as each person.** The sidebar shows only the empty menus their role may open. A JIVAH login never sees the OLIU company in the switcher.
Run 4	**Suspend one user.** Their next sign-in is refused and the attempt is in the audit trail.

PASS WHEN	Twelve logins, each scoped correctly, and the audit trail shows every sign-in. A single-company test account shows no switcher at all.
MD SEES	Their own login on the phone, the company switcher, and a list of who has access. Nothing else exists yet, and that is correct.

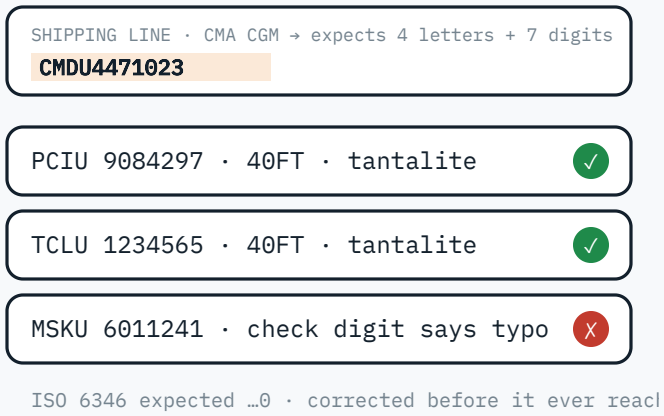
Not built yet: no bookings, no containers, no money. Resist adding a screen "while we are here".

2

Booking and container

SPINE · 2 WEEKS

The container is the unit of truth. This slice makes it impossible to type the same number twice with two meanings, which is the slip that reaches the Excel file today.



WHAT GETS BUILT

- Shipping line list with number formats
- Booking
- Container (ISO 6346 check digit, unique system-wide)
- Client
- Cargo and size
- Status: booked
- Duplicate container → MD queue stub

MENUS SWITCHED ON IN THE TENANT PORTAL

- **Shipments** → New booking
- **Shipments** → Containers: the list, not yet the board

RUNS

Run 1 **Re-enter the two October invoices as bookings.** Space Mining, 12 boxes on KOTA SETIA; Transvera, 6 boxes. Lot numbers and sizes as on the paper.

Run 2 **Type a container with a wrong check digit.** The field says what the digit should be and will not save.

Run 3 **Enter a container that already exists on another booking.** It is not rejected and not accepted: both entries go side by side to the MD.

Run 4 **Look up a box by number from the JIVAH login.** It is found, read-only, with its booking and client.

PASS WHEN Eighteen real containers are in the system with no retyping, one wrong digit was caught, and one duplicate reached the MD with both records shown.

MD SEES	A list of this month's bookings and boxes, searchable by number, from the phone.

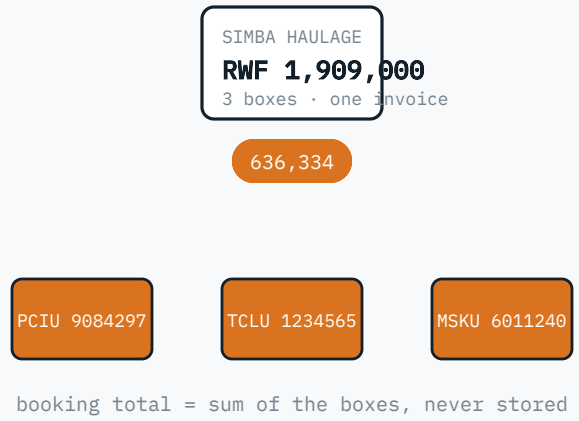
Not built yet: no costs, no stages, no legs. A booking is a list of boxes and nothing more.

3

Cost entry and the split

SPINE · 2 WEEKS

This is the slice that answers "what did this box cost". Nothing is ever entered as a bare number: booking first, then its containers appear, and a shared invoice divides across the boxes it covered.



WHAT GETS BUILT

- Cost entry (container, category, amount, currency, paid to, entered by)
- Category list
- Currency and daily rate
- Split: equal, by weight, manual
- Container cost total as a query
- Company tag from the container's side

MENUS SWITCHED ON IN THE TENANT PORTAL

- **Shipments** → Container record: cost lines
- **Shipments** → Cost & split
- **Settings** → Configuration: cost categories, currencies and rates

RUNS

Run 1 **Enter one clearing fee on one box.** It shows on that container in RWF and in USD at the day's rate.

Run 2 **Enter one truck invoice for three boxes.** Pick the booking, tick three containers, type the total once. Three cost lines appear, one per box.

Run 3 **Correct one split line.** The booking total changes because it is a sum, not a stored figure.

Run 4 **Rebuild PCIU 9084297 from the mockup.** Nine lines, USD 4,750.93, each line's rate at its own date.

PASS WHEN	Operations staff have entered a week of real costs without a spreadsheet, and the sum per container matches what the Finance Officer computes by hand for three boxes.
MD SEES	"How much did this container cost?" answered on the phone, today, for every open box.

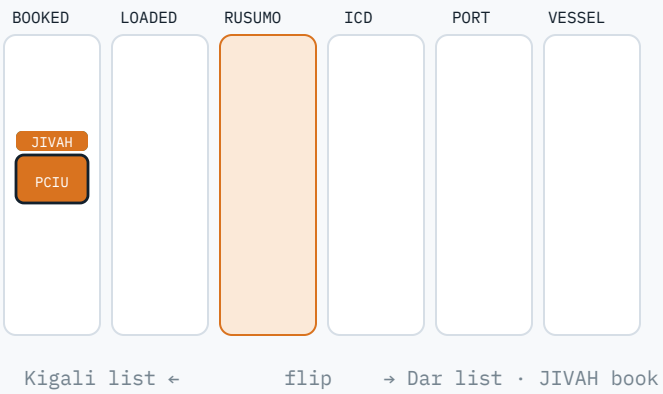
Not built yet: no payments, no approvals. A cost line records that money was spent, not that the company paid it.

4

Journey stages and the hand-over

MODULE · 3 WEEKS

The board replaces the morning round of calls. Each stage is its own fields, not a note. At Rusumo the responsible company flips from OLIU to JIVAH, and that one flip moves the box between working lists and decides which book its costs go to.



WHAT GETS BUILT

- Stage per container (8 stages, configurable)
- Board view
- Hand-over stage from set-up
- Responsible company flip
- Working list per company
- Release sign-off on pre-selected boxes
- Delay flag at ICD after N days

MENUS SWITCHED ON IN THE TENANT PORTAL

- **Shipments** → Containers: the six-column board
- **Shipments** → Journey stages
- **Shipments** → Release
- **Shipments** → Handoff to Finance
- **Shipments** → Transport leg

Transport leg stays off: a stage can be confirmed without naming the truck. Legs, breakdowns and the replacement-leg gate are their own slice after import.

RUNS

Run 1 **Move one box stage by stage.** Confirm each stage from the Kigali login until Rusumo.

Run 2 **Confirm Rusumo.** The box leaves the Kigali working list and appears on Dar's. Its flag reads JIVAH. A cost entered now lands in JIVAH's book by default.

Run 3 **Release a booking.** Boxes whose status has reached the stage are pre-selected; one still at Magerwa is not, and stays behind without anyone unticking it.

Run 4 **Leave a box at ICD past the set days.** The at-risk flag appears and a cost entered now is tagged delay-related.

PASS WHEN	Kigali and Dar staff each see only their side of a live booking, the flip happened on a real crossing, and the MD's board matches the WhatsApp group for a full week.
MD SEES	Every container in motion on one board, by stage, by company, from the phone.

Not built yet: transport legs, trucks and drivers come later as their own slice. A stage can be confirmed without a leg behind it.

5

Facilitation payments, three taps

MODULE · 2 WEEKS

The fastest screen in the system, because it is used standing up, under pressure, with no signal. A two-minute form would simply stop being used, and the control in slice 7 depends on every tip being logged.



WHAT GETS BUILT

- Phone screen, installable
- Booking remembered from last entry
- Reason buttons (configurable)
- Split across tapped boxes
- Per-tip ceiling, monthly cap, 80 % and 100 % alerts
- Payee type and MoMo / M-Pesa ID
- Offline outbox
- Complete-later screen
- Tip ledger

MENUS SWITCHED ON IN THE TENANT PORTAL

- **Field payments** → Tip entry
- **Field payments** → Complete tip
- **Settings** → Configuration: tip reasons, per-tip ceiling, monthly cap

RUNS

Run 1 **Log a tip in three taps** with the phone in flight mode. It saves. Turn data on: it syncs and appears in the ledger.

Run 2 **Tap two containers, one amount.** Two tip lines appear, split, each on its own box.

Run 3 **Reach 80 % of the monthly cap.** The MD gets an alert. At 100 % the save needs the MD's code.

Run 4 **Come back later** and attach the receipt photo and the checkpoint tag to yesterday's tip.

PASS WHEN The Field Officer has logged a week of real tips without a WhatsApp message, more than 95 % carry a payee ID, and the cap alert fired once for real.

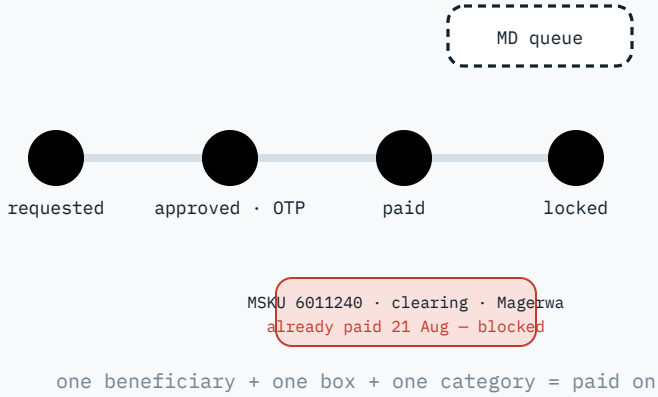
MD SEES	The tip ledger by person and by box, and cash in staff hands as a live figure.

6

Payments: request, approve, pay, lock

MODULE 3 WEEKS

Every payment is a fixed state. Nothing is paid without one-click approval, a paid order locks, and the same box plus category plus beneficiary is refused a second time and sent to the MD instead.



WHAT GETS BUILT

- Beneficiary with protected account (four-eyes on change)
- Payment request from a cost line or a tip
- Approve with MFA above threshold
- Pay and lock
- Duplicate block → MD queue
- Floats and settlement
- Office expenses through the same queue

MENUS SWITCHED ON IN THE TENANT PORTAL

- **Payments** → Needs payment: approval queue
- **Payments** → Pay — Finance Officer
- **Payments** → Beneficiaries
- **Checks** → MD queue: duplicate container, repeat payment
- **Settings** → Configuration: approval thresholds, MFA

RUNS

Run 1 **Request payment for a clearing fee.** Operations Manager approves with one click; above the threshold the code arrives at the department mailbox.

Run 2 **Pay it.** The order locks. It cannot be edited by anyone, including the MD.

Run 3 **Request the same box, fee and payee again.** It is blocked and lands in the MD queue with both orders attached.

Run 4 **Change a beneficiary's bank account.** It waits for a second person before any

payment can use it.

Run 5 **Settle a float.** The cash-in-hands figure drops the moment the settlement is approved.

PASS WHEN A month of real payments ran through the queue, every paid order is locked, one duplicate was caught before the bank, and no payment exists without an approval.

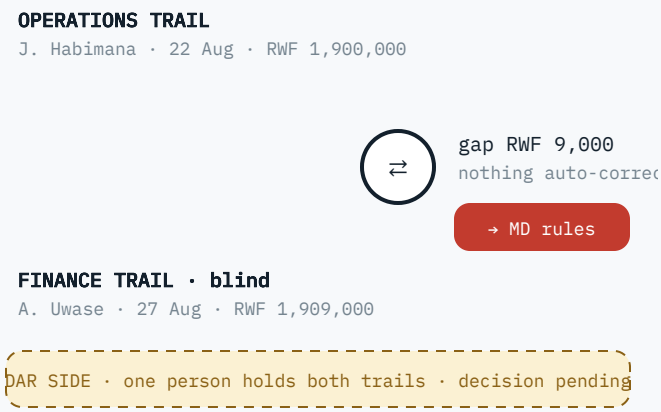
MD SEES Approvals waiting on the phone, cash out this month, and the first blocked duplicate.

7

Blind Finance entry and the compare job

DAR SIDE OPEN · MODULE · 3 WEEKS

The most important control in the system, built only now because both trails must exist first. Finance enters category, beneficiary and amount from the paper in hand, never seeing Operations' figure. A background job compares the two trails and flags a disagreement to the MD. A match is silent.



WHAT GETS BUILT

- Needs-payment queue: container only, no category, no amount
- Finance record (category, beneficiary, amount, from paper)
- Compare job every 15 min
- Flag with both figures, both authors, both times
- MD ruling as third record
- Tolerance as configuration
- "Single trail" marker for a side without blind entry

MENUS SWITCHED ON IN THE TENANT PORTAL

Payments → Pay — Finance Officer: blind view · **Checks** → Reconciliation · **Checks** → Audit trail · **Settings** → Configuration: match tolerance

Dar side: lines carry a single-trail marker and show amber on the Reconciliation tab until the MD decides.

RUNS, KIGALI PATTERN

Run 1 **Operations enters RWF 1,900,000 for leg 2.** The Finance Officer's queue shows only "PCIU 9084297 has a pending item".

Run 2 **Finance enters RWF 1,909,000 from invoice 5541.** She never saw the first figure. The compare job pairs the two lines.

Run 3 **The flag reaches the MD** with both figures. He rules Finance is right; the ruling is stored, neither record is overwritten.

Run 4 **Enter a matching pair.** Nothing happens. Silence is the healthy state.

DAR SIDE One person in Dar records both operations and finance today, so the Kigali pattern would report a match by construction. **Left open until the MD decides** between: Kigali Finance entering Dar's trail from photographed paper; the MD as Dar's second trail; big Dar costs paid by Kigali payment order with only field cash single-trail; or Dar marked single-trail and shown as such. The slice ships for Kigali either way; Dar lines carry the single-trail marker until the decision.

PASS WHEN A deliberate mismatch and a real one were both flagged to the MD within fifteen minutes, matches produced no noise for a week, and the Finance Officer confirms she never saw Operations' figure.

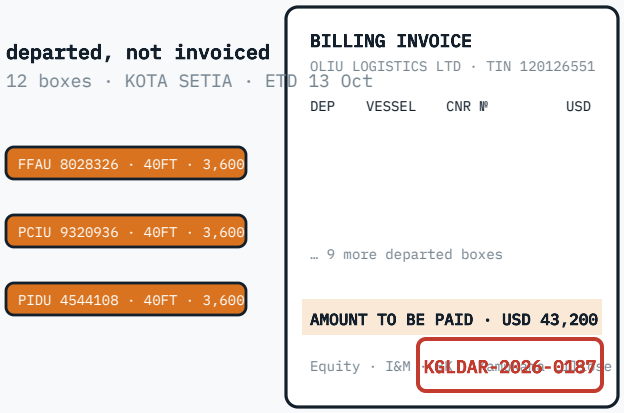
MD SEES The reconciliation flags, both trails side by side, and the first ruling he made as a third record.

8

Invoices, built by the system

MODULE · 2 WEEKS

Every column on today's paper invoice already lives in the container record. The system composes the draft from departed boxes; the Finance Officer reviews the rate and issues. Gapless number, PDF, email, no retyping.



WHAT GETS BUILT

- Rate card per client, size, route
- Draft AR invoice from departed boxes
- Issue gate: departed and costs covered
- Gapless numbering with prefix
- PDF in the current layout
- Email to client
- AP invoices
- Aged receivables, 7 / 30 / 45 day alerts

MENUS SWITCHED ON IN THE TENANT PORTAL

- **Invoices** → Invoices & clients: draft built from departed boxes, AR and AP list
- **Settings** → Configuration: invoice prefix, rate cards, client IDs
- **Settings** → Company & subscription: bank blocks and signature that print

RUNS

Run 1 **Mark the twelve Space Mining boxes departed.** A draft appears with their dep date, vessel, ETD, BL, size and lot, at USD 3,600 each.

Run 2 **Override one line's rate with a reason.** The line is flagged on the draft, not silently accepted.

Run 3 **Issue.** The number is assigned, the PDF matches KGLDAR03102026 column for column, and the email goes to the client.

Run 4 **Try to edit an issued invoice.** Refused; a credit note is the only way.

Run 5 **Let one invoice pass 30 days.** The overdue alert reaches Finance and the MD.

PASS WHEN	The system reproduced both October invoices line for line from container data, and the first real invoice went out within two working days of departure.
MD SEES	"Departed but not invoiced" at zero, clients owe us by client, and container margin for the first time.

9

Dashboard, the 07:00 digest, notifications

MODULE · 2 WEEKS

Every figure here is a live query over the slices before it. The bar for a good screen is whether it replaces a phone call. The digest is the dashboard in five lines on WhatsApp.



WHAT GETS BUILT

- Alerts-first dashboard
- KPIs: container P&L, clients owe us, cash in staff hands, cash in/out
- Notification matrix (in-app, email, SMS)
- 07:00 WhatsApp digest
- Escalation: unapproved reminds daily, overdue at 7 / 30 / 45
- Group overview for Package B

MENUS SWITCHED ON IN THE TENANT PORTAL

- **Home** → Dashboard
- **Home** → Group overview
- **Notifications** → Notifications & alerts
- **Settings** → Configuration: notification matrix, digest time
- **Home** → Ask the MIS

RUNS

- **Run 1** **Open the dashboard on the phone.** Every card is a query; change a cost and the margin moves.
- **Run 2** **Receive the digest at 07:00** for a real day and compare it with what the morning calls would have said.
- **Run 3** **Leave a payment unapproved.** The reminder repeats daily until it is ruled on.
- **Run 4** **Switch company.** The same cards for JIVAH alone, then for both.

PASS WHEN The MD has not made a morning status call for five working days and every figure on the dashboard was traced back to a line in the system.

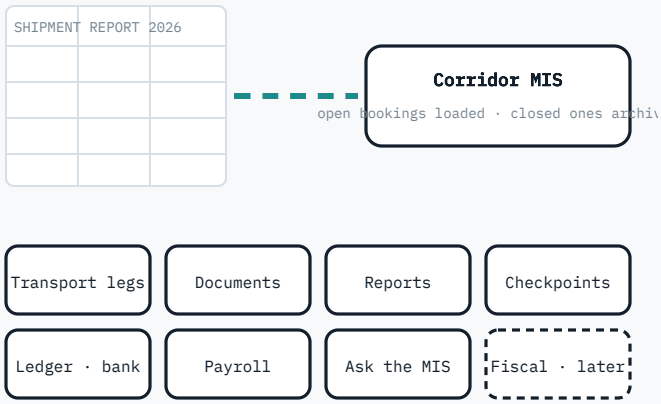
MD SEES Both companies at any hour, with what needs them at the top.

10

Import the history, then one module at a time

MODULE · 3 WEEKS + LATER SLICES

Bring the Excel history across only once the live system has proven itself. After that, every remaining feature is its own slice with its own test, in whatever order the business needs, and switching one off never touches another.



WHAT GETS BUILT

- Import wizard for SHIPMENT REPORT 2026 and FINANCE MASTER XL
- Validation report the client fixes
- Open bookings loaded, closed ones read-only archive
- Migration check signed by the MD

MENUS SWITCHED ON IN THE TENANT PORTAL

- **Settings** → Data import
- **Shipments** → Transport leg
- **Accounting** → Office expenses · Bank statement · Payroll · General ledger
- **Reports** → Reports and Excel export
- **Home** → Ask the MIS
- **Checks** → Audit trail: full filters and export

Dashed = switched on later, each as its own slice. The menu entries exist in the mockup; in the product they stay hidden for OLIU until their slice passes.

RUNS

Run 1 **Upload the raw sheet.** The wizard maps columns and lists every row that fails: bad container numbers, missing clients, duplicate bookings.

Run 2 **The client fixes the sheet** using that list. Upload again until it is clean.

Run 3 **Load.** Open and in-flight bookings appear on the board at the right stage; closed ones are an archive.

Run 4 **The MD signs the migration check:** container count, open balances and tip totals agree with the sheet.

PASS WHEN The MD signed the reconciliation and the Excel file was closed for new entries.

MD SEES The whole year in one place, and the spreadsheet retired.

THEN, EACH AS ITS OWN SLICE WITH ITS OWN TEST

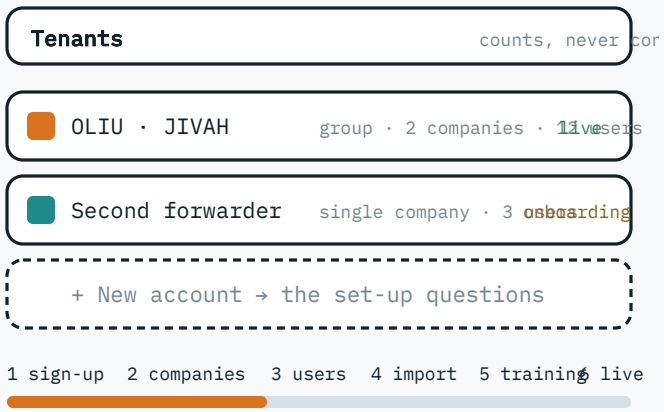
Transport legs, breakdowns, replacement-leg gate | Document checklist per cargo type | Optional checkpoint events | Reports and Excel export for the accountant | Ledger, bank matching, period lock | Payroll | Ask the MIS | Fiscal receipts, if ever asked for

11

Owner portal, stage 1: tenants and onboarding

OWNER 3 WEEKS

Not before OLIU runs its month. The owner portal earns its place the day a second forwarder is being set up, and not a day earlier. Stage 1 is only what that second onboarding needs: a list of accounts, each account's companies and users, the set-up questions as a checklist, and a log of what you did.



WHAT GETS BUILT

- Tenants list: account, companies, users, boxes this month, status
- Account page: companies & branches, users, usage counts
- Onboarding checklist: the set-up questions, import, training, pilot, live
- Your own audit log
- "Look inside" only with the tenant MD's timed approval

OWNER PORTAL SCREENS

- **Overview** → counts only
- **Tenants** → list and account page
- **Onboarding** → checklist per new account
- **Team & audit** → you, and what you did

TENANT PORTAL, SAME STAGE

- **Help** → Ask for help: a ticket with screen and last actions attached
- **Settings** → Company & subscription: platform access to your data

HOW

Run 1 **Same code, one more role.** The owner is a user whose scope is "all accounts, counts only". No new database, no second application. Every screen reads totals the tenant portal already computes.

Run 2 **Onboard the second forwarder with it.** Walk the checklist yourself: set-up questions, companies, users, their Excel through the slice 10 wizard, a training call, a three-week pilot.

Run 3 **Try to see a tenant's invoice.** You cannot. Ask their MD for a look-inside; it opens for four hours and is logged in both audit trails.

PASS WHEN A second account was created and onboarded without touching the database by hand, and the owner role reached no tenant record without an approved session.

YOU SEE Every account on one screen, where each is on the onboarding road, and nothing inside them.

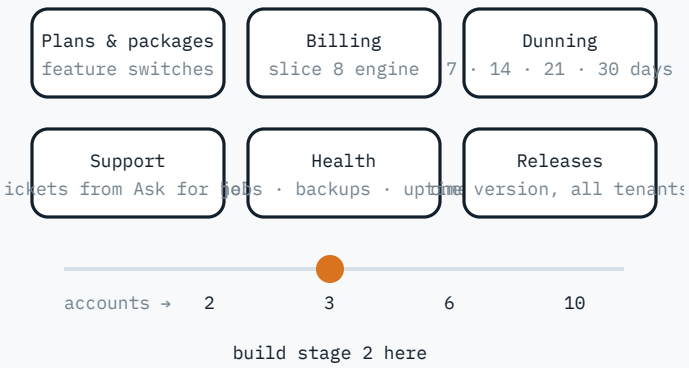
Not built yet: no plans, no subscription invoices, no support queue, no health screen. One tenant pays by bank transfer against a PDF you write; that is fine at two accounts.

12

Owner portal, stage 2: billing, support, health

OWNER · 3 WEEKS · WHEN THE THIRD ACCOUNT SIGNS

These screens replace work you can do by hand for two tenants and cannot for ten. Build them when the third account signs, not before, and reuse what exists: the subscription invoice is the slice 8 invoice engine with you as the biller, and the feature switches are the settings slice 1 already stores.



WHAT GETS BUILT

- Packages and feature switches per account
- Subscription invoice on the 1st, through the slice 8 engine
- Dunning: reminders, banner, read-only, suspend; nothing deleted
- Support queue fed by the tenant's Ask for help
- Health: jobs, backups, restore test, uptime
- Releases: one version, notes the tenant reads

OWNER PORTAL SCREENS

- **Plans & packages** → switches per account
- **Billing** → invoices, payments, dunning
- **Support** → tickets, severity, response times
- **Health & regions** → jobs and backups
- **Releases** → version and feature switches

HOW

Run 1 **Bill yourself first.** Generate the 1 November run for the three accounts; check each PDF against the package. Match a bank transfer to one invoice by hand.

Run 2 **Let one go unpaid on purpose.** Watch the reminders fire at 7 and 14 days, the banner

at 21, read-only at 30. Confirm the tenant can still export everything.

Run 3 **Raise a ticket from inside a tenant.** It arrives with the screen and the last twenty audit events attached. Answer it from the queue.

Run 4 **Switch a feature off for one account.** Payroll disappears from that tenant's sidebar; its records stay in reports. No other tenant notices.

PASS WHEN A month's subscription invoices went out without you, one unpaid account went read-only and came back on payment, and a feature switch changed one tenant's sidebar and nothing else.

YOU SEE Who pays, who is late, who needs help, and whether last night's backup ran, in one morning glance, like the MD's digest but for the platform.

Still later, only if asked: a second hosting region, fiscal receipts, tax-authority links, shipping-line APIs.

The road at a glance

Slice	Weeks	One developer, one slice at a time
1 · Accounts, companies, roles	1–2	
2 · Booking and container	3–4	
3 · Cost entry and split	5–6	
4 · Journey stages and hand-over	7–9	
5 · Tip entry	10–11	
6 · Payments	12–14	
7 · Blind entry and compare · Dar open	15–17	
8 · Invoices	18–19	
9 · Dashboard and digest	20–21	
10 · Import, then later modules	22–24 →	
11 · Owner portal stage 1	25–27	
12 · Owner portal stage 2	28–30	

About six months for one developer to a system OLIU runs its month on, then six more weeks for the owner portal, timed to the second and third accounts rather than to the calendar. Three developers in parallel can shorten the tenant slices only after slice 3, because slices 1 to 3 are the spine everything else hangs on; a realistic parallel plan is slices 4, 5, 6 together, then 7, 8, 9 together, which lands OLIU in late January. Each week-count includes the week of live use that closes the slice. The proposal's 14 December date is reachable only by shipping slices 1, 2, 3, 6 and 8 first and the rest from February; that is a decision for the MD, not a default.

Why this order and not the proposal's

- **Spine first.** Account, container, cost. Without those three nothing else can be tested on real data.
- **Value early.** After slice 3, six weeks in, OLIU already knows what each box cost. That is the first thing the owner asked for.
- **The control last among the core.** Blind entry and compare need both trails to exist, so it sits at slice 7, not slice 1 where its importance would tempt you to put it.
- **Invoices after payments.** The client invoice waits until the box's own costs are covered, so payments must exist before the issue gate can be real.
- **Import late.** Migrating history into a system that is still changing shape is wasted work. Migrate once, into a system that has held a live month.
- **Owner portal last, in two stages.** With one tenant it is a form you fill yourself. It becomes a screen when the second account is onboarded and a billing system when the third signs. Building it earlier would be building for customers who do not exist.

What stays open while we build

- **Dar blind entry.** Slice 7 ships for Kigali. Dar lines carry a single-trail marker until the MD chooses one of the four arrangements. No code depends on the choice.
- **Who bills the client: decided, OLIU.** The August analysis said JIVAH; the MD has confirmed OLIU, as the October invoices show. Set at account creation in slice 1; the invoice slice reads it.
- **Reconciliation tolerance.** Exact match by default, a band for conversion drift. A setting in slice 7.
- **Escort with the cargo, or dispatcher.** Decides how hard slice 5's offline mode has to work. Build offline anyway; it costs little.

Corridor MIS · build road · drawn from the August 2026 systems analysis §11, the proposal's phases, and the two mockups · menu names are the SaaS mockup's

Pricing deliberately left out